Project canvas

Framework RME — risk, mitigation and evidence. AI system governance · Applied Research · PUCPR.

PROBLEM · THE THREE MISSING LINKS

Gap 1 · no link

The organization lists risks and lists controls, but never records which control addresses which risk, nor who is accountable.

Gap 2 · effectiveness and cost unmeasured

There is no proof that the mitigation is in place, and no comparison between the cost expected and the cost actually spent.

Gap 3 · no post-deployment observation

The risk assessment ages: nothing brings the mitigation back for review when the system changes, fails or causes harm.

COMPONENTS OF THE ARTIFACT

C1 · Link model

Ties risk, mitigation, owner, evidence and cost, with two status dimensions: progress and verification.

Gap 1 Gap 2

C2 · Mitigation catalogue

40 measures curated from Saeri et al. (2025), under CC BY 4.0, with target risk, expected evidence, cost and uncertainty assigned by the group.

Gap 1

C3 · Monitoring protocol

An operational definition of adverse event, five reassessment triggers and a review interval per risk tier: 90, 180 and 365 days.

Gap 3

C4 · Working prototype

Laravel 13 · React 19 · Inertia · PostgreSQL 18. An append-only audit trail and a traceability report per system.

Gap 2 Gap 3

Actors

- User — authenticates
- Technical team — registers, links, proves and verifies
- Production monitor — records events and changes
- Product manager — reassesses reverted links
- Compliance and legal — monitors and exports
- Independent auditor — reads the trail in the report
- Scheduler — reverts overdue links, with no author

Success criteria and results

- **OE1** inter-annotator agreement on the target risk of mitigations: Krippendorff alpha with MASI distance of **0.527** (95% CI 0.389–0.646), below the 0.667 minimum and reported as is.
- **OE2** the model holds mitigations tied to several risks with no change to the architecture: **met**.
- **OE3** the adverse event definition validated by experts: **not verified**.
- **OE6** content validity of the artifact by the panel: **not verified**.

Scope and limitations

- The expert panel was planned and **not run**: the protocol, the bank of 21 items and the CVR calculation are ready for a future run.
- The catalogue is limited to 40 mitigations, scoped to the organization that **deploys and operates** AI systems, not to frontier model developers.
- Permissive authorization in the prototype: access roles are left for future work.
- In production, reversal on review due depends on a scheduler on the server.